

Generalizing Beyond the Cutoff: A Within-Study Comparison of Regression Discontinuity and Difference-in-Differences Estimates of a Hybrid Tutoring Program¹

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September 18, 2026

Abstract

Programs that assign services to students scoring below an eligibility threshold are often evaluated with regression discontinuity designs. These estimates are internally valid but apply only to students at the margin of eligibility, who are rarely the students furthest behind. We use a two-year panel of Renaissance Star reading and mathematics scores from a large urban district to carry a causal estimate from the cutoff to the full population assigned to Targeted Skills Instruction (TSI), a virtual tutoring program integrated with Edmentum's Exact Path platform. We estimate the intent-to-treat effect of assignment four times, changing one element of the design at each step: a sharp regression discontinuity in endline levels; a difference-in-discontinuities that removes a pre-existing baseline gap; a two-period inverse probability-weighted difference-in-differences for all assigned students; and its six-period event-study extension. An exact algebraic equivalence joins the second and third estimates, so moving from the cutoff to the full population changes the sample, not the estimator. In reading, the effect rises from 0.11 standard deviations at the cutoff to 0.18 under the aligned near-cutoff estimator and 0.25 for the full assigned population. In mathematics, the cutoff benchmark is null once the baseline gap is removed, and the negative full-population estimates rely on one identification strategy and attenuate under alternative weighting. We read the mathematics evidence as ruling out benefit rather than demonstrating harm.

¹ The research reported here was supported by the Institute of Education Sciences, U.S. Department of Education, through Grant R305B200005 to the University of Virginia. This work was made possible through a research partnership with Edmentum, which provided the data used in this study. We are grateful to Julie Cohen, Daphna Bassok, and Brendan Bartanen for helpful comments on previous versions of this work. The views expressed are those of the authors and do not necessarily represent those of the Institute, the U.S. Department of Education, or Edmentum. Any errors are our own.

Introduction

High-impact tutoring is among the most effective interventions for accelerating student learning, with meta-analyses of randomized trials reporting effects of 0.30 to 0.40 standard deviations when tutoring is frequent, embedded in the school day, and aligned with core instruction (Nickow, Oreopoulos, & Quan, 2024; Robinson, Kraft, Loeb, & Schueler, 2021). It is also expensive, and districts adopting it at scale cannot serve every student. Seats are rationed, and the usual rationing device is a diagnostic cutscore that assigns the program to students scoring below a threshold, as in the remedial and summer-school programs long studied in the program-evaluation literature (Jacob & Lefgren, 2004; Matsudaira, 2008) and in the tutoring program we evaluate here. The cutscore creates a bind. Cutoff-based assignment is what makes credible causal inference possible without randomization, because a regression discontinuity can compare students just below the threshold to students just above it. But the effect it identifies is the effect for students at the margin of eligibility, and a targeted program's margin is precisely where its theory of change matters least. The students tutoring exists to serve sit far below the threshold, where the discontinuity is silent.

This mismatch between the best-identified estimand and the policy-relevant one is well recognized (Wing & Cook, 2013), and an econometric literature has developed methods for extrapolating RD estimates away from the cutoff (Angrist & Rokkanen, 2015; Dong & Lewbel, 2015; Cattaneo, Keele, Titiunik, & Vazquez-Bare, 2021). For high-impact tutoring the mismatch is consequential for two reasons. First, every decision a district actually faces, whether to adopt the program, how many seats to fund, where to set the cutscore, and whom to prioritize when seats are scarce, turns on the effect for the full assigned population or on how the effect varies with distance below the threshold, not on the effect at the margin. Second, tutoring that targets

diagnosed skill gaps is built on the premise that it is most valuable for the students with the largest gaps. If that premise holds, effects should grow with distance below the cutoff, and a cutoff-local evaluation will systematically understate the program's value for its target population. Whether it holds is the empirical question this study is designed to answer.

We study these questions in an evaluation of Targeted Skills Instruction (TSI), a virtual tutoring program that integrates with Exact Path, a personalized learning platform developed by Edmentum. The combination pairs the scalability of educational technology with the motivational support and scaffolding of live tutoring (Cheung & Slavin, 2013; Escueta, Nickow, Oreopoulos, & Quan, 2020). A prior fuzzy regression discontinuity evaluation (Li, Ostrow, & Dray, 2026) found a complier effect of 0.51 standard deviations on state ELA scores for students near the cutoff. We extend that evaluation with panel data. The district administers Renaissance Star interim assessments in fall, winter, and spring, so we observe each student's achievement four times before TSI began and twice after, in Winter and Spring 2024–25. We use the panel for two purposes. First, the pre-intervention measurements let us test the assumptions of the level RD: whether a gap at the cutoff was already present before assignment, and whether that gap was stable over time. Second, the full panel supports a difference-in-differences anchored to the RD estimate at the cutoff, which identifies the effect for the full population of assigned students and traces how the effect unfolds over the intervention year. We estimate both in reading and mathematics.

We estimate a single intent-to-treat effect, the effect of assignment to TSI, four times, changing one element of the design at each step. A sharp regression discontinuity (RD) at the diagnostic cutoff identifies the effect of assignment on endline achievement for students at the margin. Because endline levels embed any gap that existed before the intervention, we

decompose that discontinuity exactly into a baseline discontinuity and a discontinuity in growth, and the growth discontinuity is our cutoff-local benchmark. An inverse probability-weighted difference-in-differences (IPW-DiD), with weights estimated from pre-intervention achievement trajectories, then identifies the effect for the full assigned population, first over two periods and then as a six-period event study. Restricted to the RD bandwidth, the two-period IPW-DiD can be made algebraically identical to the RD estimator, so any difference between the near-cutoff and full-population estimates reflects the population averaged over, not a change of design.

The study makes two contributions. The first is methodological. Within-study comparisons evaluate a quasi-experimental design by placing it beside a more credible benchmark applied to the same target population (LaLonde, 1986; Cook, Shadish, & Wong, 2008; Wing & Cook, 2013). We extend this tradition from comparison to construction. Rather than placing an RD and a DiD side by side, we align the two estimators exactly, on a common sample, scale, outcome, functional form, and weighting, so that their point estimates coincide algebraically at the cutoff, and we then use that anchored estimator to identify the effect for the population the cutoff cannot reach. Unlike RD extrapolation methods that stay within the RD framework under a conditional ignorability assumption (Angrist & Rokkanen, 2015), we switch to a second design whose identifying assumption is directly testable in the pre-intervention panel, and we verify the switch with an exact equivalence at the cutoff. Each step changes exactly one element (the outcome, the estimator, the population, the panel length), so the change in the estimate at each step has one named source ([Figure 1](#)). The second contribution is substantive. Districts deciding whether to adopt a program, how to ration its seats, and where to set its cutoff care about the effect for the full population the program serves, above all the students furthest below the threshold. The four estimates show that this differs from the effect at the margin: in

reading the estimated effect of assignment rises from 0.11 standard deviations at the cutoff to 0.25 for the full assigned population, while in mathematics a null cutoff benchmark sits beside negative, less robust full-population estimates. An evaluation confined to the margin would have missed both patterns.

Background

Prior Evidence on Tutoring and Personalized Learning Technology

High-impact tutoring raises achievement by 0.30 to 0.40 standard deviations in the meta-analytic literature but remains hard to scale. Traditional one-on-one and small-group models require trained personnel, coordination with schools, and sustained funding, and the large effects in the meta-analytic literature come disproportionately from small, tightly controlled programs. Reweighting the experimental evidence toward the conditions under which tutoring is actually delivered at scale in U.S. schools, Kraft, Schueler, and Falken (2024) find expected effects of roughly 0.10 to 0.15 standard deviations, one-third to one-half the canonical pooled estimates, with program scale the strongest predictor of the decline. Recent experiments nonetheless identify design features that relax the constraints. Staffing can be broadened: the Saga tutoring model, delivered by recent college graduates and paraprofessionals with targeted training, raised math achievement by 0.16 to 0.37 standard deviations in Chicago high schools (Guryan et al., 2023). Delivery can move online when dosage and tutor quality are maintained: intensive online tutoring by qualified tutors produced gains of roughly 0.20 to 0.26 standard deviations for disadvantaged secondary students in Spain and Italy (Carlana & La Ferrara, 2024; Gortazar, Hupkau, & Roldán-Monés, 2024), and an experiment that randomly assigned the same tutors to in-person or remote early-literacy tutoring found no modality difference (Hashim, Miles, & Croke, 2025). Virtual delivery is not uniformly effective, however: a virtual model for

kindergarten through second grade produced average reading gains of only 0.05 standard deviations, with larger effects for one-on-one and for lower-performing students (Robinson, Pollard, Novicoff, White, & Loeb, 2025). Dosage, group size, and tutor quality, rather than the virtual format itself, appear to be the binding constraints.

A second route to scale substitutes technology for part of the tutor's time. A 4:1 model in which two students worked with a tutor while two used adaptive software, alternating daily, raised math achievement by 0.19 standard deviations among ninth graders in Chicago and New York (Bhatt et al., 2024), and interim results from the multi-district Personalized Learning Initiative, which randomized more than 17,000 students, found that a “sustainable” model doubling group size and substituting software for a portion of tutor time was as effective as full high-dosage tutoring at roughly half the cost, with virtual tutors as effective as in-person ones (Bhatt et al., 2025). Evidence on personalized learning technology on its own points the same way. Adaptive software that adjusts instruction to students' measured skill levels shows some of the most consistent positive effects among education technology interventions, whereas hardware alone yields little (Escueta et al., 2020); the strongest causal evidence, a 0.37 standard deviation math gain in 4.5 months among Delhi middle schoolers, came from adaptive software embedded in small-group facilitator support (Muralidharan, Singh, & Ganimian, 2019). The recurring lesson across successes and failures is that personalized learning technology performs best when a human instructional layer surrounds it.

Targeted Skills Instruction

Targeted Skills Instruction (TSI) combines these two ingredients. Exact Path, Edmentum's personalized learning platform, begins with a diagnostic assessment and generates an individualized learning path of lessons, practice, and mastery checks aligned to state

standards. TSI adds live, small-group virtual tutoring sessions embedded in the school day. Sessions last 30 to 45 minutes, follow a structured format of direct instruction, guided practice, independent work, and formative assessment, and occur multiple times per week; tutors use students' Exact Path diagnostic and progress data to select focal skills. Within this multi-tiered arrangement Exact Path serves students at all achievement levels, while TSI delivers more intensive support to students one to two grade levels behind. Edmentum recommends assigning TSI to students below the 25th percentile on the Exact Path diagnostic, though teachers ultimately determine participation. The program's theory of change holds that tutoring enhances the technology by sustaining engagement, reinforcing understanding, and supporting persistence through difficult material. An Edmentum-conducted fuzzy regression discontinuity evaluation of the same implementation and student population studied here found that TSI participation raised state ELA scores by 0.51 standard deviations among compliers near the cutoff (Li et al., 2026); the present study asks how the effect of assignment generalizes from that margin to the full assigned population, in both subjects.

Eligibility was determined by the Exact Path Insights Score, a diagnostic screener on a vertically scaled 500 to 1,500 point scale that permits comparison across grades. Separate reading and mathematics cutoffs were set for each grade at the 25th percentile of the score distribution: 875, 926, and 985 in reading and 814, 880, and 965 in mathematics for grades 3, 4, and 6. The screener was administered in early September of the 2024–25 school year and the first TSI session began on September 30. Eligibility was fixed by this single administration and never updated. Many eligible students nonetheless did not receive TSI, for reasons including chronic absenteeism, lack of notification, login problems, teacher non-compliance, and

scheduling conflicts, and some students above the cutoff received it, presumably at teacher discretion.

Research Questions

Our question concerns the effect of assignment to TSI on reading and mathematics achievement among students in schools that use Edmentum products, compared with non-assigned students in the same schools. Rather than posing it as a single question, we specify four estimands that differ in the population for which the effect is identified and in the design that identifies it:

- (1) What is the effect of assignment to TSI on Spring 2024–25 achievement *levels* for students at the eligibility cutoff, identified by a sharp regression discontinuity?
- (2) What is the effect of assignment to TSI on achievement *growth* for students at the eligibility cutoff, identified by a difference-in-discontinuities design that removes the baseline discontinuity embedded in the first question?
- (3) What is the effect of assignment to TSI on achievement growth for *all* assigned students, identified by a two-period inverse probability-weighted difference-in-differences?
- (4) What is the trajectory of the effect of assignment to TSI for all assigned students across the intervention year, identified by a six-period inverse probability-weighted event study?

The questions are ordered so that each differs from its predecessor in exactly one respect;

Figure 1 presents the four steps, with each arrow naming the single element that changes between them. Moving from the first question to the second changes the outcome, from endline levels to Spring-to-Spring growth, which removes any gap at the cutoff that predates the intervention. Moving from the second to the third changes the population, from students at the margin of eligibility to the full assigned population; this move is made interpretable by the

bridge described in the next section, under which the two-period IPW difference-in-differences restricted to the RD bandwidth is algebraically identical to a sharp RD on the change score.

Moving from the third to the fourth changes the panel, from the two comparison springs to all six testing periods, which traces the timing of the effect and connects the sequence to a conventional event-study presentation. Each design carries its own identifying assumptions, and the next section pairs each step with the checks that test them.

Data, Measures, and Sample

Data and Measures

The study uses third-, fourth-, and sixth-grade students from 30 schools in a large urban district in the Northeastern United States, the grades in which TSI was piloted during 2024–25. Twenty-three schools implemented Edmentum’s Exact Path and TSI that year and seven did not. The 23 Edmentum schools form the confirmatory sample for every step, which compares students assigned to TSI with non-assigned students in the same schools. The seven non-Edmentum schools support a check on within-school spillovers, described with the event study below. Three data sources are linked: district administrative records with student demographics; a two-year panel of Renaissance Star reading and mathematics scores covering 2023–24 and 2024–25; and Edmentum records identifying Exact Path users, TSI participants, and each student’s Exact Path Insights diagnostic score. From these we built one analytic dataset per subject.

The outcomes are the Renaissance Star Reading and Math assessments, computer-adaptive tests that are vertically scaled, so that scores across grades sit on a single continuous scale and individual growth can be tracked over time. The district administers them in fall, winter, and spring for formative purposes (universal screening, progress monitoring, and goal

setting), not for accountability, so they measure growth rather than attainment of a proficiency criterion. They are therefore a more proximal indicator of achievement growth than a state test, and their three-times-yearly administration supplies the four pre-intervention measurements on which the designs below depend.

Demographic covariates are binary, time-invariant indicators for gender (male), race and ethnicity (White, Hispanic, Native American, Asian, Black, Pacific Islander, and two or more races), multilingual learner status, economic disadvantage, non-English home language, learning disability, and homelessness, together with grade indicators. They play three roles: they describe the samples, they are tested for balance at the cutoff as a check on the RD, and the cutoff-local models adjust for them, entering the vector Z_i linearly with a common coefficient on both sides of the cutoff (White and third grade are the omitted categories). The full-population models use them in none of these ways: the propensity model excludes them by design, and student fixed effects absorb all time-invariant characteristics.

The diagnostic score defines both treatment-side variables. The assignment indicator equals one if the student scored at or below the grade- and subject-specific cutoff on the single September 2024 administration; it is the treatment variable in the difference-in-differences and event-study models. The discontinuity designs use the same score and cutoff to form the running variable, the student's assignment depth, defined in the next section. Both variables reflect eligibility under the cutoff rule rather than program receipt, which is what allows an intent-to-treat framework to be shared across all four steps.

Sample Descriptives

Tables 1 and 2 report sample characteristics for reading and mathematics: all students in the two-year panel at Edmentum schools with a diagnostic score, before the missing-data

exclusions and common-support restrictions that define the estimation samples. Each table gives means for assigned and non-assigned students (Columns 1 and 2) and, because the sequence begins at the cutoff, for the assigned and non-assigned students whose diagnostic scores fall within the RD bandwidth used throughout (Columns 3 and 4). The reading sample contains 1,015 assigned and 1,598 non-assigned students, of whom 521 and 526 lie within the bandwidth; the mathematics sample contains 973 assigned and 1,669 non-assigned students, of whom 659 and 936 lie within the bandwidth. Consistent with the program's targeting, assigned students differ from non-assigned students on most characteristics: they are more likely to be Hispanic or Black and less likely to be White or Asian, and more likely to be multilingual learners, economically disadvantaged, learning disabled, and homeless, with the largest imbalances on multilingual and learning disability status in both subjects. Their raw pre-intervention achievement trajectories are correspondingly lower, by roughly 135 scale-score points in reading in Fall 2023–24, and flatter than those of non-assigned students, who rise more steeply between Fall and Spring 2023–24 before leveling off.

These descriptives motivate each piece of the design. The large level differences and diverging raw pre-trends confirm that the full assigned and comparison groups are not directly comparable, so any full-population estimate needs a design that removes stable differences and balances trajectories; the weighted difference-in-differences supplies both. The within-bandwidth columns show what the cutoff produces: near the threshold, assigned and non-assigned students are far more alike. In reading, the mean absolute standardized difference across the sixteen characteristics falls from 0.20 in the full sample (Column 5) to 0.08 within the bandwidth (Column 6), and the number of significant imbalances falls from twelve to two; in mathematics the declines are from 0.21 to 0.11 and from eleven to seven. This local comparability is why the

sequence begins at the cutoff. The same columns also quantify the cost of staying there. Only about half of assigned readers (521 of 1,015) and two-thirds of assigned mathematics students (659 of 973) score within the bandwidth, and the bandwidth slice is systematically less disadvantaged than the assigned population as a whole: among assigned readers the multilingual share falls from 39% in the full group to 29% within the bandwidth, and mean Spring 2023–24 achievement rises from roughly 825 to 862 scale-score points, closing about a quarter of the gap to non-assigned students. The closer a student’s diagnostic score sits to the cutoff, the less that student resembles the furthest-behind population the program targets, which is why the estimates cannot stop at the cutoff.

Non-compliance with the assignment rule is predominantly one-sided. In reading, 460 of the 1,015 assigned students (45%) did not receive TSI, and in mathematics 439 of 973 (45%) did not; in the other direction, only 35 of 1,598 non-assigned readers (2%) and 95 of 1,669 non-assigned mathematics students (6%) received TSI. The consequences for interpretation are taken up in the next section.

Identification Strategy

Our identification strategy moves from the population where identification is most credible to the population of greatest policy interest, and the pre-intervention panel is what makes each move testable. It begins with a sharp regression discontinuity at the diagnostic cutoff and ends with a six-period event study for the full assigned population; [Table 7](#) assembles the four estimates.

The approach follows the within-study comparison tradition, in which estimates from a quasi-experimental design are evaluated against a benchmark from a more credible design applied to the same target population. LaLonde (1986) launched the tradition by showing that

standard econometric adjustments failed to recover the experimental estimate of a job-training program's effect, and Cook, Shadish, and Wong (2008) codified the conditions under which observational designs reproduce experimental benchmarks: a common target population, a common estimand, and aligned specifications. Our method is closest to Wing and Cook (2013), who benchmark pretest-supplemented discontinuity designs against an experiment and show that pre-intervention measures of the outcome make the RD's assumptions transparent and testable, improve its precision, and support estimates for populations away from the cutoff. We build on exactly this result. Our study contains no experiment, so the sharp RD, paired with four pre-intervention outcome measures, serves as the benchmark. It qualifies for that role on the grounds the within-study tradition requires: it identifies from a known assignment mechanism rather than from assumptions about unobserved counterfactual trajectories, its identifying assumptions are weaker and more local than the DiD's, and the pre-intervention panel lets us check their observable implications, including whether any discontinuity at the cutoff was already present before assignment. The difference-in-differences then extends the benchmark beyond the cutoff, which matters because of what Wing and Cook identify as the sharp RD's decisive weakness relative to an experiment: its estimand is confined to the narrow subpopulation at the cutoff.

Because of non-compliance, every step estimates an intent-to-treat effect of assignment rather than an effect of receipt. Roughly 45% of assigned students did not take up TSI, so the ITT is attenuated relative to the effect among compliers, and every estimate below should be read as a conservative lower bound on the effect of receiving TSI. The ITT is also the more policy-relevant quantity: non-compliance is expected in practice, and the ITT reflects the effect of assigning students and making the program available. Because our RD is specified in reduced

form on assignment rather than receipt, it targets the same estimand as the DiD, unlike the prior fuzzy RD evaluation (Li et al., 2026), which estimated a complier effect of receipt.

The cutoff-local designs share a set of conventions. Let s_i be student i 's diagnostic score and c_g the grade- and subject-specific cutoff; students scoring at or below the cutoff are assigned, $T_i = 1\{s_i \leq c_g\}$. We express the rule through the running variable $r_i = c_g - s_i$, the student's *assignment depth*: the number of scale-score points by which the diagnostic score falls at or below the cutoff, positive below the cutoff and negative above it, so that $T_i = 1\{r_i \geq 0\}$ and larger values identify the students the program targets most intensively. The bandwidth h is selected once per subject as the MSE-optimal bandwidth of the covariate-adjusted sharp RD on Spring 2024–25 levels, $h = 65.80$ scale-score points in reading and $h = 72.40$ in mathematics, and is held fixed in every cutoff-local model. Those models use the balanced near-cutoff sample: students in Edmentum schools with $|r_i| \leq h$ and non-missing Star scores in both Spring 2023–24, the reference period before any TSI activity, and Spring 2024–25, the endline, giving 912 students in reading and 1,358 in mathematics, with outcomes standardized to the Spring 2023–24 mean and standard deviation of non-assigned students within the bandwidth. The full-population models use the broader generalization sample described below.

The Cutoff Benchmark

The first step is the conventional sharp RD: the discontinuity in Spring 2024–25 achievement levels at the cutoff. We regress the standardized endline score on the running variable, the assignment indicator, their interaction, and the covariates,

$$y_{i,\text{Spring } 25} = \alpha + \phi_1 r_i + \tau_1 T_i + \phi_2 (T_i \times r_i) + Z_i' \psi + \varepsilon_i,$$

by weighted least squares with triangular kernel weights $w_i = \max(0, 1 - |r_i|/h)$. Here ϕ_1 is the slope of the outcome in the running variable on the non-assigned side, ϕ_2 the change in that

slope on the assigned side, ψ the covariate coefficients, and τ_1 , the coefficient of interest, the discontinuity in levels at the cutoff, reported with a robust bias-corrected confidence interval (Calonico, Cattaneo, & Titiunik, 2014; covariates enter as in Calonico, Cattaneo, Farrell, & Titiunik, 2019). Identification rests on smoothness of potential outcomes at the cutoff: absent treatment, students just below and just above the threshold would have indistinguishable outcomes. We probe this assumption three ways. A manipulation test asks whether the density of the running variable jumps at the cutoff, as it would if scores had been manipulated to place students on a preferred side (McCrary, 2008), using the local polynomial density estimator of Cattaneo, Jansson, and Ma (2020); manipulation is institutionally implausible, since eligibility was fixed by a single September diagnostic before the rule was salient, but the test formalizes the claim. Covariate balance tests estimate a sharp RD of each covariate at the fixed bandwidth; significant discontinuities motivate the covariate adjustment in the level models. And binned RD plots (Figures 2 and 3) show the raw discontinuities in both the endline level and the change score.

The level RD conflates two quantities. Because the RD estimator at a fixed bandwidth is linear in the outcome, the level discontinuity decomposes exactly into

$$\hat{\tau}_{RQ1} = \hat{\tau}_{\text{base}} + \hat{\tau}_{RQ2},$$

where τ_{base} is the discontinuity in Spring 2023–24 levels, a full year before assignment, and τ_2 is the discontinuity in the Spring-to-Spring change score $\Delta y_i = y_{i, \text{Spring } 25} - y_{i, \text{Spring } 24}$. Both are estimated with the level specification above and only the outcome replaced (Table 3). The baseline discontinuity is a pure diagnostic: under the smoothness logic of the level RD it should be zero, and any nonzero value measures how much of the level benchmark reflects a gap that predates the intervention. The growth discontinuity is the second step, a difference-in-

discontinuities (Grembi, Nannicini, & Troiano, 2016; Picchetti, Pinto, & Shinoki, 2024): the effect of assignment on growth at the cutoff, purged of any pre-existing gap. Because covariates are time-invariant, differencing removes their level effects; they are retained so that the three rows of [Table 3](#) share one specification and decompose exactly, and without them the growth discontinuity moves by 0.01 standard deviations or less in either subject.

The difference-in-discontinuities identifies the effect only if the baseline discontinuity is time-invariant: the gap at the cutoff would have carried forward unchanged into Spring 2024–25 absent the intervention (Picchetti et al., 2024). Following Wing and Cook (2013), we test this condition in the pre-intervention panel three ways. We estimate the level discontinuity separately in each of the six testing periods; if the confounding discontinuity is time-invariant, the pre-period estimates should be stable. We pool the four pre-intervention periods and jointly test whether the shape of the outcome-by-running-variable function and the size of the jump changed across periods; and because that test cannot reach the intervention year, a companion test restricted to non-assigned students, the only students untreated in every period, pools Spring 2023–24 with Spring 2024–25 and asks whether their untreated outcome function changed between the two springs beyond a level shift. And we plot student-level change scores against the running variable to check that they are flat through the cutoff. Two supplementary diagnostics re-estimate the period-by-period discontinuities on the subsample observed in every pre-period, to separate real gap dynamics from changing composition, and test directly for a monotone drift in the discontinuity with a single cutoff-by-time interaction; the drift’s implication for the growth discontinuity is derived as a bound on the cutoff benchmark (supplementary materials).

Generalizing Beyond the Cutoff

The third and fourth estimates extend the estimand to the full assigned population with an inverse probability-weighted difference-in-differences: student fixed effects difference out stable characteristics, testing-period fixed effects absorb common time shocks, and inverse probability weights reweight the comparison group so that its pre-intervention trajectory matches the assigned group's. The identifying assumption is conditional parallel trends: absent TSI, the two groups' achievement would have evolved similarly, conditional on the pre-intervention trajectories on which the weights are estimated. The central threat is the program's targeting. Students one to two grade levels behind may follow systematically different trajectories, faster under regression to the mean or slower under persistent disadvantage, so unaddressed trajectory differences could bias the estimate in either direction. The weights address this threat directly, and the four pre-intervention periods make the resulting balance verifiable.

The generalization sample comprises all assigned and non-assigned students in the 23 Edmentum schools, with outcomes standardized to that sample's Spring 2023–24 control mean and standard deviation (a slightly different basis from the cutoff-local models', because control-group dispersion differs inside and outside the bandwidth; [Table 7](#)'s note marks where the scale changes). To construct the weights we fit, separately by subject, a logistic regression of assignment on the three consecutive pre-intervention gain scores $g_{i,k} = y_{i,k+1} - y_{i,k}$, the changes between adjacent testing periods before the intervention. We condition on gains rather than levels for two reasons. The threat to identification is differential pre-intervention growth, which the gains capture directly, while baseline levels are absorbed by the student fixed effects. And because assignment is a deterministic function of the diagnostic score, which prior levels proxy closely, a propensity model in levels largely reconstructs the assignment rule: fitted propensities

pile up near zero and one and the overlap on which reweighting depends collapses, a pattern the robustness checks document empirically. Demographic characteristics are excluded because conditioning on them did not improve, and sometimes degraded, trajectory balance, and the fixed effects already absorb them. From the fitted propensity p_i we form weights targeting the effect among the treated, here the assigned: assigned students receive a weight of one and comparison students the odds of assignment, $p_i / (1 - p_i)$, so the reweighted comparison group mimics the assigned group's trajectory. Before weighting we restrict both groups to the region of common support, trimming students whose propensities fall outside the 1st and 99th percentiles of the assigned distribution, so that the estimates do not extrapolate beyond the support of the assigned population (imputation of missing pre-period gains and alternative trimming rules: supplementary materials).

In reading, for example, the sample begins with 1,012 assigned and 1,598 non-assigned students, the common-support restriction leaves 991 and 1,593, and weighting leaves 991 assigned students and an effective comparison sample of roughly 977. Mean control weights are close to one and tightly distributed, so no small group of comparison students exerts undue influence; propensity distributions overlap across the common-support region; and, after weighting, the assigned and comparison trajectories track one another closely across all four pre-intervention periods, supporting the conditional parallel trends assumption on which the full-population estimates rest.

The Bridge

Restricting the DiD to the RD bandwidth puts both designs on the same students, but it does not make them the same estimator. Even on identical students, the two headline estimates differ on four mechanical margins that have nothing to do with their identifying assumptions.

The RD is estimated on endline levels; the DiD is estimated on the change from one spring to the next. The RD fits a separate line on each side of the cutoff; the DiD compares two group means. The RD weights students by a triangular kernel that fades with distance from the cutoff; the DiD weights them by the inverse probability weights. And the RD reports a bias-corrected estimate, while the DiD reports an ordinary regression coefficient. Placed side by side, then, the two numbers can differ for any of these reasons, and a reader cannot tell whether a gap between them reflects disagreement about the effect or a difference in arithmetic.

The bridge removes the arithmetic. Both estimators are weighted least squares coefficients. The conventional sharp RD at a fixed bandwidth is the coefficient on the assignment indicator in a kernel-weighted regression of the outcome on polynomials in the running variable fitted separately on each side of the cutoff. The two-period DiD is also a weighted regression: with two periods, the student fixed effects reduce the panel model to a regression of each student's change score on the treatment indicator. When the two regressions are the same regression, their coefficients coincide exactly. That happens once each of the four margins is aligned: the RD is estimated on the change score rather than the level; it uses a local-constant fit, which is a difference in mean gains, the same quantity the DiD estimates; it uses a uniform kernel, which weights every student in the bandwidth equally, with the IPW weights carried across as unit weights; and the comparison uses the conventional RD estimate at the fixed bandwidth on a common balanced sample with a single standardization. Under these alignments the two-period IPW DiD on the balanced near-cutoff sample,

$$y_{it} = \gamma_i + \delta_t + \tau_{br}(T_i \times \text{Post}_t) + \varepsilon_{it},$$

where γ_i and δ_t are student and testing-period fixed effects, Post_t marks Spring 2024–25, and τ_{br} is the difference in mean gains between assigned and non-assigned students, estimated

with the ATT weights and student-clustered standard errors, is algebraically identical to a sharp RD on the change score with a local-constant fit, a uniform kernel, and the IPW weights as unit weights. We verified the identity, and each intermediate estimator pair in the alignment sequence, numerically: the point estimates agree to machine precision, with differences on the order of 10^{-15} (derivation: supplementary materials).

Aligning the point estimates does not align the standard errors, and the difference is informative. DiD inference treats the estimand as a weighted difference in mean gains and applies a cluster-robust standard error; RD inference targets the limit of the outcome regression at the cutoff and, in its robust bias-corrected form, additionally prices the error of the local polynomial approximation near the boundary, so its intervals are mechanically wider for the same coefficient. [Table 4](#) therefore reports the single bridge estimate twice, under each framework. When the estimate is significant under one and not the other, the divergence reflects the different inferential standards the designs apply, not conflicting evidence about the effect.

Widening the Sample, Lengthening the Panel

The bridge fixes the estimator at the cutoff. The third step holds the estimator fixed and widens the population. The same two-period IPW DiD, with τ_3 in place of τ_{br} , is estimated on the generalization sample, restricted to common support and balanced on the two comparison springs ([Table 5](#)). Because the estimator is identical, the difference between τ_{br} and τ_3 reflects the population over which the effect is averaged, students at the margin versus the full assigned range, and not any change in design. If effects were homogeneous along the running variable the two would coincide up to sampling error; where they diverge, the divergence measures how the effect for the full assigned population differs from the effect at the margin. [Figure 5](#) visualizes exactly this comparison.

The fourth and final step extends the panel to all six testing periods,

$$y_{it} = \sum_{j=-3, j \neq -1}^2 \beta_j (T_i \times D_{jt}) + \gamma_i + \delta_t + \varepsilon_{it},$$

where D_{jt} indicates that period t falls in event time j , measured relative to the introduction of TSI, and the model is estimated by weighted least squares with the same weights and student-clustered standard errors. TSI launched after the Fall 2024–25 testing window, $j=0$. We normalize to $j=-1$ (Spring 2023–24) rather than to this final pre-period because Fall 2024–25 scores were measured concurrently with the eligibility screener, so differences at that point could partly reflect the screening process itself; Spring 2023–24 is a clean baseline and aligns the event study with the change-score construction of every other step. Event time runs from $j=-3$ (Fall 2023–24) to $j=2$ (Spring 2024–25); β_1 and β_2 are the post-intervention effects, we define $\tau_4 \equiv \beta_2$ as the fourth and final estimate, and we also report the pooled post-intervention effect, the average of the two, with delta-method standard errors. The pre-period coefficients are a falsification test: under no anticipation and correct specification they should be near zero and jointly insignificant (Miller, 2023). The event study inherits the conditional parallel trends assumption; its extra pre-periods are what make it partially testable. Its panel is unbalanced, the one respect in which its sample differs from the two-period sample: students missing the Spring 2023–24 score contribute through other periods but drop out of the change score.

Two further checks accompany the full-population estimates. The design assumes each student’s outcomes depend only on their own assignment, which would fail if TSI affected non-assigned classmates, for instance by reallocating teacher attention. We test this with a cross-school comparison in which the comparison group is students in the seven schools that never adopted Edmentum products, who cannot be reached by any within-school spillover, re-estimating the event study with the same weighting machinery and pairing it with two

falsification tests that estimate the “effect” of not being assigned and of attending an Edmentum school at all against the same no-exposure baseline; the first doubles as a direct spillover test. And we subject the weighted model to robustness checks that vary the sample restrictions (balanced panels), the weighting model (levels in place of gains, with and without demographics), and the common-support rule. Both sets of results are summarized with the main findings (full analyses: supplementary materials).

Results

We present results in the order of the four steps: the cutoff, the bridge, the full population and the intervention year, and the four estimates together. In brief, in reading the effect of assignment is positive at every step and strengthens as the estimand widens from the cutoff to the full assigned population, from 0.11 standard deviations at the margin, to 0.18 under the aligned near-cutoff estimator, to 0.25 for all assigned students, and disagreement between the designs is confined to inferential standards rather than point estimates. In mathematics the sequence is diagnostic: the negative level discontinuity predates the intervention, the cutoff growth benchmark is null once that gap is differenced out, and the full-population estimates are negative, significant in spring but sensitive to the weighting model. All estimates are intent-to-treat effects of assignment in standard-deviation units and are conservative lower bounds on the effect of receiving TSI given the one-sided non-compliance documented above.

At the Cutoff

The sharp RD passes its design checks. The manipulation test finds no discontinuity in the density of the running variable in either subject (robust bias-corrected $p = .149$ in reading, $p = .424$ in mathematics), consistent with an eligibility rule fixed by a single September diagnostic. Covariate balance at the cutoff is consistent with a valid design: most characteristics show no

discontinuity, none is imbalanced in mathematics, and the two modest discontinuities in reading, learning disability status (coefficient = 0.074, $p = .038$) and the fourth-grade indicator (coefficient = -0.098 , $p = .056$), motivate the covariate adjustment carried by every cutoff-local model. The binned plots (Figures 2 and 3) show the raw discontinuities: in reading, a visible jump in Spring 2024–25 levels and a smaller jump in the change score; in mathematics, level and change-score patterns that are close to smooth through the threshold.

The level discontinuity is the first row of Table 3. In reading, students just below the cutoff scored 0.31 standard deviations higher in Spring 2024–25 than students just above it (95% CI [0.17, 0.81], $p = .002$), a large local effect taken at face value. In mathematics the level discontinuity is negative and not significant (-0.17 , 95% CI [-0.49 , 0.12], $p = .23$). The second and third rows decompose each level estimate exactly into its baseline and growth components, and in both subjects the level estimate turns out to be a misleading summary of the intervention year. In reading, the 0.31 splits into a 0.20 standard deviation gap already present in Spring 2023–24 (95% CI [-0.15 , 0.63], $p = .23$) and a 0.11 discontinuity in Spring-to-Spring growth (95% CI [-0.11 , 0.62], $p = .17$). In mathematics the decomposition is starker: the -0.17 consists almost entirely of a statistically significant -0.16 baseline discontinuity (95% CI [-0.58 , -0.06], $p = .018$), leaving a growth discontinuity of -0.01 (95% CI [-0.14 , 0.40], $p = .34$). Students just below the mathematics cutoff were already scoring lower a full year before assignment, which violates the smoothness logic of the level RD; a posttest-only RD would have read the negative level estimate as harm. The growth discontinuities, 0.11 in reading and -0.01 in mathematics, are the cutoff benchmarks for everything that follows.

The time-invariance tests certify the growth benchmark against every alternative except one, and that exception makes the benchmarks conservative. In mathematics the pre-period

discontinuity is consistently negative and stable, from -0.13 to -0.19 across the four pre-intervention periods, exactly the pattern the assumption requires; the joint tests do not reject (shape $p = .124$, jump $p = .585$, untreated-function $p = .168$); and the change-score plots are essentially flat. In reading no pre-period discontinuity is individually significant and none of the joint tests reject (shape $p = .756$, jump $p = .246$, untreated-function $p = .882$), but the point estimates drift from roughly 0.25 standard deviations in fall and winter 2023–24 to 0.05–0.09 by spring. The drift is not compositional: on the balanced subsample observed in every pre-period, the reading estimates still fall from 0.31–0.33 to 0.13 to 0.05. And the targeted trend test detects a small downward drift in both subjects, -0.06 standard deviations per testing period in reading ($p = .019$) and -0.05 in mathematics ($p < .001$), unchanged when the running-variable polynomial is allowed to differ by side. The near-identity of the two slopes points to a common panel artifact, such as mean reversion on the shared outcome scale, rather than subject-specific gap dynamics. Either way the decomposition signs the consequence: because the growth discontinuity subtracts the Spring 2023–24 gap as the stand-in for the counterfactual Spring 2024–25 gap, a gap still drifting toward zero implies any bias is downward. Taken at face value, the drift implies the reading benchmark of 0.11 understates the cutoff effect by roughly 0.12 standard deviations, and the mathematics null shades toward the positive rather than concealing harm (supplementary materials). Estimated without covariates the growth discontinuities are essentially unchanged (0.12 in reading, -0.01 in mathematics).

The Bridge

Table 4 reports the bridge: the two-period IPW DiD on the balanced near-cutoff sample, algebraically identical to a sharp RD on the change score with a local-constant fit, a uniform kernel, and the IPW weights as unit weights. Its two rows differ only in the inferential

framework applied to a single number. In reading the bridge estimate is 0.18 standard deviations. Under cluster-robust DiD inference it is significant ($SE = 0.09$, $p = .039$, 95% CI [0.01, 0.35]); under robust bias-corrected RD inference the identical point estimate is not (95% CI [-0.19, 0.29], $p = .70$). Neighboring change-score RD specifications behave the same way: the local-linear estimate at the fixed bandwidth is positive but insignificant without weights (0.12, $p = .19$), with the inverse probability weights (0.11, $p = .19$), or with a freshly selected MSE-optimal bandwidth (0.33, $p = .19$). We read this as agreement rather than conflict. Every aligned specification points to a positive reading effect of roughly 0.1 to 0.2 standard deviations near the cutoff, and the significance discrepancy reflects the RD's more demanding inferential standard, not contrary evidence.

In mathematics the bridge estimate is -0.12 and not significant under either framework (cluster-robust $p = .11$, 95% CI [-0.27, 0.03]; robust bias-corrected $p = .31$, 95% CI [-0.31, 0.10]). The gap between the null benchmark and the negative bridge estimate is itself informative, because the two numbers summarize the same students on the same outcome and only the weighting changes. The growth discontinuity evaluates the jump at the boundary, with a triangular kernel concentrating weight on students immediately around the cutoff; the bridge replaces this with an IPW-weighted mean contrast over the full bandwidth window. That the estimate moves from -0.01 to -0.12 indicates that the mathematics growth gap is near zero at the threshold itself but widens with assignment depth even within the bandwidth. This is a caution for the steps that follow. The estimator step moves the estimate in both subjects, but in reading the movement preserves the conclusion, since both weightings show a positive effect of similar size, whereas in mathematics it creates one, turning a boundary null into a window-average negative. The negative DiD-family estimates emerge exactly as weight shifts from the immediate

cutoff toward deeper-assigned students, the region where a depth-concentrated effect and depth-correlated trajectory divergence cannot be separately identified.

Beyond the Cutoff

Table 5 applies the same two-period estimator to the generalization sample. In reading, the effect of assignment for the full assigned population is 0.25 standard deviations (95% CI [0.13, 0.37], $p < .001$, $N = 2,289$). Because the estimator is identical to the bridge specification, the movement from 0.18 at the bandwidth to 0.25 for the full population is a difference in estimand, not design: students far below the threshold, whom only the DiD can reach, gained more than students at the margin. In mathematics the full-population estimate is -0.12 and statistically significant (95% CI $[-0.23, -0.01]$, $p = .038$, $N = 2,211$), the first significant negative estimate in the sequence. Read causally, as the reading gradient is, it would indicate harm concentrated among the lowest-scoring students. Two considerations counsel caution. The mathematics estimate has no anchor: the extensively tested benchmark at the cutoff is null, so the negative result lives entirely in the deep-assigned region where the design distinguishes least well between treatment effects and trajectory divergence. And its robustness is uneven, holding under the sample-side checks but attenuating to null under alternative weighting models.

The six-period event study (Table 6, Figure 4) traces the timing. In reading there is no evidence of differential pre-trends ($F(3, 2583) = 1.01$, $p = 0.39$). The pooled post-intervention effect is 0.20 standard deviations ($SE = 0.04$, $p < 0.001$), present in both post-periods and growing: 0.16 in Winter 2024–25 ($SE = 0.04$, $p < 0.001$) and 0.23 in Spring 2024–25 ($SE = 0.05$, $p < 0.001$), a pattern consistent with a cumulative benefit over the intervention year. The spring coefficient is the fourth estimate and sits close to the two-period estimate (0.23 versus 0.25); the small wedge reflects panel balance, since students missing the Spring 2023–24 score contribute

to the event study but not to the change score. In mathematics there is again no evidence of differential pre-trends ($F(3, 2566) = 1.07, p = 0.36$). The pooled effect is -0.08 ($SE = 0.04, p = 0.054$), combining a null in Winter 2024–25 ($\beta = -0.03, SE = 0.04, p = 0.48$) with a significant negative in Spring 2024–25 ($\beta = -0.12, SE = 0.05, p < 0.01$) that matches the two-period estimate exactly. We give this spring-concentrated pattern limited confirmatory weight: a negative effect that appears only in the final period, with a null at midyear, lacks the cumulative profile that lends the reading results their coherence, and is equally consistent with late-emerging harm and with residual trajectory divergence that accumulates over the panel.

The robustness checks reinforce this asymmetry. The reading effect is stable across every viable specification, with pooled estimates of 0.17 to 0.21 standard deviations that are significant in all cases: 0.20 and 0.21 under the two balanced-panel restrictions, 0.17 ($p = .007$) when demographic covariates enter the weighting model, and 0.18 to 0.20 across the alternative common-support rules. The mathematics estimate survives the panel and trimming variations (-0.09 and -0.06 under the balanced panels; -0.07 to -0.10 across trimming rules) but attenuates from -0.08 ($p = .054$) to -0.02 ($p = .71$) in the pooled estimate, and from -0.12 ($p < .01$) to -0.04 (n.s.) in Spring 2024–25, once demographic covariates enter the weighting model. Replacing gain scores with achievement levels in the propensity model does not yield a viable alternative; it reconstructs the assignment rule, with maximum control weights rising from below 8 to as high as 380 and erratic, imprecise estimates, confirming the overlap argument for the gain-score model. The cross-school check likewise supports the within-school design. Relative to students in non-Edmentum schools, beyond the reach of any within-school spillover, assigned readers gained 0.13 standard deviations (pooled, $SE = 0.03, p < 0.001$), agreeing with the within-school estimate in sign and significance; the direct spillover test is null (non-assigned Edmentum

students versus non-Edmentum students: pooled -0.03 , $SE = 0.02$, $p = 0.14$), as is the Edmentum-school falsification test (pooled 0.03 , $SE = 0.02$, $p = 0.16$). In mathematics the cross-school estimate is null (pooled -0.05 , $SE = 0.04$, $p = 0.14$) but rests on a comparison with a marginal pre-trend violation ($F(3, 1982) = 2.51$, $p = 0.057$), so it denies the within-school negative independent corroboration without cleanly certifying its absence; both mathematics falsification tests are null (supplementary materials).

The Four Estimates Together

[Table 7](#) assembles the four estimates, and [Figure 5](#) draws the central comparison. In reading, the sequence reads as a generalization result: 0.31 at the cutoff in levels; 0.20 of that gap already present at baseline; 0.11 in growth at the cutoff; 0.18 under the aligned estimator on the same near-cutoff sample; 0.25 for the full assigned population; and 0.23 in the six-period event study. [Figure 5](#) shows why the near-cutoff and full-population estimates differ: binned mean gains rise steadily with distance below the threshold, so the local fits separate by only 0.12 standard deviations at the cutoff while the assigned mean sits 0.25 standard deviations above the IPW counterfactual across the full assigned range. The gap between the steps is a difference in estimands, not a disagreement between designs, and a cutoff-based evaluation alone would have understated the program's value for the students it most needs to serve. Two cautions attach to the gradient and are taken up in the Discussion: variation in gains along the running variable can reflect differential counterfactual growth as well as heterogeneous effects, and the cutoff anchor itself may be conservative, in which case part of the rise across the steps reflects the anchor rather than depth heterogeneity.

In mathematics, the sequence reads as a diagnosis. The level discontinuity (-0.17) is an artifact of a significant pre-existing baseline gap (-0.16); the growth discontinuity is null

(-0.01), and its time-invariance assumption survives every joint test, with the one detected deviation, a small downward drift common to both subjects, signed to shade the null toward the positive rather than conceal harm; and the full-population estimates are negative (-0.12 in both the two-period and six-period designs), significant in spring but not winter, and sensitive to the weighting model. The mathematics evidence rules out benefit. DiD-family point estimates are consistently negative and occasionally significant, but they share one identification strategy, attenuate to null under alternative weighting models, and find no corroboration in the extensively tested null benchmark at the cutoff. The data cannot distinguish a small negative effect concentrated among the lowest-scoring students from residual trajectory confounding the weights fail to remove.

Discussion

The central question motivating this study was one of generalization. The prior causal evidence on TSI came from a fuzzy regression discontinuity design and therefore spoke only to students at the margin of eligibility, compliers scoring near the 25th-percentile diagnostic cutoff (Li et al., 2026). But TSI is not a program for marginal students. It is assigned to every student at or below the cutoff, a population that extends far below the threshold, and the students it most needs to serve are those furthest from the margin. The cutoff-local estimand, however internally valid, is rarely the one of direct policy interest (Wing & Cook, 2013). Our answer to this problem is to estimate the effect four times, changing one element at each step. Rather than choosing between the RD's credibility and the DiD's reach, we ordered the designs so that each step changes exactly one element: the outcome, the estimator, the population, the panel length. We joined the two designs with the bridge, an exact algebraic equivalence: on the near-cutoff sample, the two-period IPW difference-in-differences *is* a sharp RD on the change score, to

machine precision. Because the estimator does not change while the sample widens, the change in the estimate from step to step measures how the effect itself varies across populations, not how the specification varies across designs.

In reading, the result is visible in [Figure 5](#). The effect of assignment generalizes beyond the cutoff and strengthens: 0.11 standard deviations in growth at the eligibility margin, 0.18 under the aligned estimator on the same near-cutoff students, 0.25 for the full assigned population, with binned gains rising steadily with assignment depth. Read causally, the students who began furthest below the threshold, one to two grade levels behind and the population TSI's theory of change targets, gained the most. A cutoff-based evaluation alone would have understated the program's value for its target group. This gradient warrants care. Variation in gains along the running variable can reflect differential counterfactual growth, such as regression to the mean among low scorers, as well as heterogeneous treatment effects, and the design identifies the average effect of assignment, not its distribution across assignment depth. Still, the pattern is what one would expect if tutoring aligned to diagnosed skill gaps is most productive for the students with the largest gaps. The cross-school check, with its null direct spillover test and falsification tests, supports both the within-school design and the attribution of the reading gains to TSI assignment rather than to Edmentum exposure or school-level selection.

A second qualification attaches to the gradient's lower anchor. The time-invariance diagnostics admit the possibility that the cutoff benchmark of 0.11 is conservative by roughly the size of the detected pre-period drift; corrected for that drift, the cutoff effect would approach the full-population estimate, and the movement from 0.11 to 0.25 would then reflect the correction of a downward-biased anchor rather than effect heterogeneity. The gradient claim therefore rests most safely on the comparison the drift cannot touch: the identical IPW difference-in-differences

estimator, whose identification runs through weighted parallel trends rather than the subtraction of a baseline gap, yields 0.18 on the near-cutoff sample and 0.25 on the full assigned population, a same-design movement of 0.07 standard deviations that is evidence of a real, though more modest, depth gradient. Under either reading the policy conclusion is unchanged, since a cutoff-local evaluation understates the program's value for its full assigned population in both scenarios; but the strong form of the heterogeneity claim, that gains are concentrated among the furthest-behind students, should be held with the anchor's uncertainty in mind (supplementary materials).

The reading gradient has direct consequences for two decisions districts make about a cutoff-assigned program. The first is whom to serve when seats are scarce. If effects grow with distance below the threshold, then filling seats near the margin buys the smallest gains available, and prioritizing the lowest-scoring eligible students concentrates the program where its returns appear largest. The second is whether to expand eligibility. A proposal to raise the cutoff above the 25th percentile should be judged by the effect for the students it would newly admit, and those students sit at the margin, where the estimated effect is roughly 0.1 to 0.2 standard deviations, not by the full-population average of 0.25. The estimates show that the two quantities differ substantively, and that using the average to forecast the expansion would overstate its value.

Mathematics tells a different story, and the four estimates are what constrain its telling. The mathematics evidence rules out benefit. DiD-family point estimates are consistently negative and occasionally significant, but they share one identification strategy, attenuate to null under alternative weighting models, and find no corroboration in the extensively tested null benchmark at the cutoff. The data cannot distinguish a small negative effect concentrated among the lowest-

scoring students from residual trajectory confounding the weights fail to remove. The asymmetry in how we read the two subjects requires a defense, because the arithmetic is symmetric: in both subjects the estimate moves away from the cutoff benchmark as weight shifts toward deeper-assigned students, and read causally, the same gradient logic that carries the reading conclusion would read the mathematics pattern as harm concentrated among the lowest-scoring students. Two features of the evidence justify treating the cases differently. First, the two sequences are anchored differently. In reading, the cutoff benchmark is itself positive, and the population extension strengthens an effect that already exists at the margin; the difference-in-differences extends the RD. In mathematics, the extensively tested benchmark is null, and the negative estimates are created, not extended, by the reweighting: the negative sign appears exactly as weight moves from the immediate cutoff toward deeper-assigned students, the region where a depth-concentrated effect and depth-correlated trajectory divergence cannot be separately identified. Second, the two gradients differ in robustness. The reading gradient survives every weighting specification, the balanced-panel and trimming variations, and the cross-school falsification tests, and its cutoff anchor carries only a qualification that runs in its favor, since any continuation of the pre-period drift would make 0.11 an understatement, not an artifact. The mathematics estimates attenuate to null under alternative weighting models and find no corroboration in the cross-school comparison, which is itself compromised by a pre-trends violation. A gradient produced by the estimator's weighting that does not survive changes to it is what depth-correlated confounding would look like. A gradient anchored at the cutoff that survives every check is what a real effect would look like.

For the program, the practical conclusion does not depend on resolving this ambiguity. Either mathematics tutoring as delivered produced no benefit, or it produced a small harm

concentrated among the students the program most intensively targets; under neither reading does the mathematics model warrant scaling in its current form, and the contrast with reading, produced by the same platform, tutors, and delivery model, points the diagnosis toward the mathematics-specific content and its integration rather than the hybrid model itself. For the same reason, the cutoff-placement implications the reading results support have no mathematics counterpart: decisions about where to set a threshold presuppose a credible effect gradient along the running variable, and in mathematics the estimated gradient is exactly the quantity the design cannot certify.

More broadly, the study illustrates what a panel of interim assessments gives an evaluator of a cutoff-assigned intervention. The pre-intervention outcomes made every identifying assumption in the four designs testable to some degree: the density and balance checks tested the RD, the stacked discontinuities and functional-form tests tested the difference-in-discontinuities, and the pre-trend falsification tests tested the event study. They also exposed a threat no single design could have detected alone, the mathematics baseline discontinuity, which a posttest-only RD would have read as evidence of harm and which the decomposition instead reveals to predate the intervention. The construction generalizes beyond this application. Any intervention assigned by a cutoff and observed in a panel supports the same construction: a sharp RD anchored at the threshold, a difference-in-discontinuities that removes pre-existing gaps, an exact bridge to a reweighted difference-in-differences, and a population extension whose movement is interpretable because the estimator is held fixed. Where randomization is not feasible, designs sequenced in this way can broaden the population for which effects are identified while each design tests the assumptions of the next.

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Appendix: Sources of Reported Results

Every table, figure, and number reported in the paper traces to an analysis script and an output file in the project folder (Table A1). Scripts live in 04_analysis_code/ and their outputs in 06_output/. Exhibit numbers in the paper differ from the numbers embedded in the output filenames; each row's output filename identifies the artifact. Derivations, the full time-invariance and drift diagnostics, the robustness tables, and the cross-school analysis are in the supplementary materials.

Table A1. Sources of the results reported in the paper.

Result	Reported in	Script(s)	Output file(s)
Schematic of the four steps	Figure 1	create_estimand_ladder_figures.py --short	short_figure_1_four_steps_schematic.png
Sample characteristics, full sample and within bandwidth (counts 1,015/1,598, 521/526, 973/1,669, 659/936; standardized differences 0.20→0.08, 0.21→0.11; imbalance counts; multilingual share 39%→29%; Spring 2023–24 means 825→862)	Sample Descriptives; Tables 1–2	02_descriptives.do	table_1_balancetable_read.rtf, table_2_balancetable_math.rtf
Raw pre-intervention trajectories (≈135-point reading gap, flatter assigned-group trend)	Sample Descriptives	02_descriptives.do	figure_2_raw_trajectories_read.png, figure_3_raw_trajectories_math.png
Non-compliance counts (460 of 1,015; 439 of 973; 35 of 1,598; 95 of 1,669)	Sample Descriptives	02_descriptives.do	table_3_compliance_within_school.rtf
Bandwidths (65.80, 72.40) and near-cutoff sample sizes (912, 1,358)	Identification Strategy	03_rd_assumption_checks.do, 04_rd_ladder.do	rd_assumption_checks.log, rd_ladder.log
Manipulation test ($p = .149$, $.424$) and covariate balance at the cutoff (0.074 , $p = .038$; -0.098 , $p = .056$)	At the Cutoff	03_rd_assumption_checks.do	table_5_rd_density_test.rtf, table_4_rd_covariate_balance.rtf
Binned RD plots	Figures 2–3	03_rd_assumption_checks.do	figure_5_rd_plots_read.png, figure_6_rd_plots_math.png
Level discontinuity, baseline discontinuity, growth discontinuity, with and without covariates	At the Cutoff; Table 3	04_rd_ladder.do	table_8_rd_decomposition.rtf, rd_ladder.log, ladder_rungs_rd.dta
Bridge estimate under both inference frameworks;	The Bridge; Table 4	04_rd_ladder.do	table_9_aligned_did

Result	Reported in	Script(s)	Output file(s)
neighboring change-score RD specifications; machine- precision equivalence			_rd.rtf, rd_ladder.log
Period-by-period discontinuities, joint shape/jump tests, untreated- function test	At the Cutoff	08_wing_cook_validi ty_tests.do	table_6_stacked_rd_ read.rtf, table_7_stacked_rd_ math.rtf, wing_cook_validity_ tests.log, figure_7_changescor e_diagnostics_read. png, figure_8_changescor e_diagnostics_math. png
Balanced-panel discontinuities and drift trend test (-0.06 , $p = .019$; -0.05 , $p < .001$; implied ≈ 0.12 bias)	At the Cutoff; Discussion	12_balanced_panel_s tability_check.do	table_D1_stacked_rd_ balanced.rtf, table_D2_drift_tren d_test.rtf, balanced_panel_stab ility_check.log
Generalization-sample construction, weights, common support (1,012/1,598 $\rightarrow$ 991/1,593 $\rightarrow$ 991/ ≈ 977), weight summaries, overlap, weighted pre-trends	Generalizing Beyond the Cutoff	00_matching_program .do, 01_build_analysis_d atasets.do, 02_descriptives.do, 09_consort.do, create_consort_funne l.py	consort_counts.csv, figure_4_consort_fu nnel.png, table_10_ipw_summar y_stats.rtf, figure_9_pretrends_ read.png, figure_10_pretrends_ math.png, figure_11_propensit yscoreoverlap_read. png, figure_12_propensit yscoreoverlap_math. png
Full-population two-period DiD (0.25 , $N = 2,289$; -0.12 , $N = 2,211$)	Beyond the Cutoff; Table 5	05_two_period_did_f ull_sample.do	table_11_two_period_ did_full_sample.rtf, two_period_did_full_ sample.log, ladder_rung_1c.dta
Six-period event study (pre- trend F tests, pooled and period effects)	Beyond the Cutoff; Table 6 ; Figure 4	06_eventstudy.do	table_12_eventstudy .rtf, figure_13_eventstud y.png, ladder_rung_1d.dta
Cross-school comparison and falsification tests (0.13 ; -0.03 ; 0.03 ; -0.05 ; $F(3$, $1982) = 2.51$)	Beyond the Cutoff	01_build_analysis_d atasets.do, 06_eventstudy.do	table_B5_eventstudy_ crossschool_read.r tf, table_B6_eventstudy_ crossschool_math.r tf
Robustness checks (balanced panels, weighting models, trimming rules; maximum control weights)	Beyond the Cutoff	10_did_robustness_c hecks.do	table_A1_did_robust ness_read.rtf, table_A2_did_robust ness_math.rtf
Generalization figure (0.12)	Figure 5 ; The Four Estimates	07_generalization_f	figure_14_generaliz

Result	Reported in	Script(s)	Output file(s)
separation at the cutoff; 0.25 across the assigned range)	Together; Discussion	figure.do	ation.png, generalization_figure.log
Summary of the four estimates	Table 7	11_ladder_summary_table.do	table_13_estimand_ladder.rtf, ladder_summary_table.log

Tables

Table 1

Reading Sample Characteristics, Full Sample and Within the RD Bandwidth

	(1) TSI Assigned	(2) Non- Assigned	(3) TSI Assigned, Within Bandwidth	(4) Non-Assigned, Within Bandwidth	(5) STD Difference (1 v 2)	(6) STD Difference (3 v 4)
3rd Grade	0.42	0.47	0.47	0.49	0.09 (0.02)	0.03 (0.60)
4th Grade	0.43	0.40	0.40	0.38	-0.06 (0.16)	-0.03 (0.61)
6th Grade	0.15	0.13	0.13	0.13	-0.05 (0.20)	0.00 (0.95)
Male	0.54	0.47	0.49	0.50	-0.15 (0.00)	0.02 (0.80)
Race White	0.24	0.35	0.26	0.29	0.26 (0.00)	0.06 (0.29)
Race Hispanic	0.47	0.39	0.41	0.42	-0.18 (0.00)	0.03 (0.64)
Race Native American	0.01	0.00	0.01	0.00	-0.02 (0.65)	-0.02 (0.65)
Race Asian	0.06	0.15	0.09	0.08	0.23 (0.00)	-0.03 (0.47)
Race Black	0.33	0.27	0.34	0.38	-0.13	0.09

					(0.00)	(0.20)
Race Pacific Islander	0.01	0.01	0.02	0.01	-0.05	-0.09
					(0.22)	(0.10)
Race Two or More Races	0.01	0.02	0.01	0.02	0.10	0.05
					(0.01)	(0.35)
Multilingual	0.39	0.12	0.28	0.20	-0.69	-0.22
					(0.00)	(0.00)
Economically Disadvantaged	0.83	0.74	0.82	0.86	-0.20	0.07
					(0.00)	(0.15)
Home Language is Not English	0.51	0.41	0.45	0.40	-0.21	-0.10
					(0.00)	(0.10)
Learning Disability	0.31	0.08	0.27	0.13	-0.61	-0.36
					(0.00)	(0.00)
Homeless	0.06	0.02	0.05	0.04	-0.21	-0.04
					(0.00)	(0.62)
Student N	1,015	1,598	521	526		
School N	23	23	23	23		

Note. Column 1 presents means for students assigned to TSI in schools that adopted Edmentum products during the 2024–25 school year. Column 2 presents means for students not assigned to TSI in the same Edmentum schools. Columns 3 and

4 restrict columns 1 and 2 to students whose diagnostic scores fall within the covariate-adjusted MSE-optimal bandwidth of the Spring 2024–25 level RD (65.80 scale-score points in reading, 72.40 in mathematics) around the grade-specific eligibility cutoff. Columns 5 and 6 present standardized mean differences — between the full assigned and non-assigned groups (5) and between the assigned and non-assigned groups within the bandwidth (6) — with p-values in parentheses. Standardized differences are calculated as the difference in group means divided by the pooled standard deviation of the full sample, so the within-bandwidth differences are on the same scale as the full-sample differences.

Table 2

Mathematics Sample Characteristics, Full Sample and Within the RD Bandwidth

	(1) TSI Assign ed	(2) Non- Assign ed	(3) TSI Assigned, Within Bandwidth	(4) Non-Assigned, Within Bandwidth	(5) STD Difference (1 v 2)	(6) STD Difference (3 v 4)
3rd Grade	0.40	0.47	0.46	0.48	0.15 (0.00)	0.03 (0.51)
4th Grade	0.41	0.42	0.41	0.41	0.02 (0.60)	0.00 (0.95)
6th Grade	0.19	0.11	0.13	0.11	-0.25 (0.00)	-0.05 (0.35)
Male	0.50	0.49	0.48	0.47	-0.03 (0.51)	-0.03 (0.53)
Race White	0.21	0.36	0.22	0.30	0.33 (0.00)	0.18 (0.00)
Race Hispanic	0.48	0.39	0.47	0.46	-0.17 (0.00)	-0.02 (0.66)

Race Native American	0.01	0.00	0.00	0.00	-0.02 (0.56)	0.00 (0.95)
Race Asian	0.05	0.16	0.05	0.09	0.28 (0.00)	0.09 (0.01)
Race Black	0.37	0.24	0.37	0.30	-0.31 (0.00)	-0.17 (0.00)
Race Pacific Islander	0.01	0.01	0.01	0.01	0.01 (0.74)	0.06 (0.14)
Race Two or More Races	0.01	0.02	0.01	0.02	0.10 (0.01)	0.07 (0.10)
Multilingual	0.34	0.18	0.33	0.25	-0.38 (0.00)	-0.21 (0.00)
Economically Disadvantaged	0.85	0.72	0.86	0.79	-0.29 (0.00)	-0.16 (0.00)
Home Language is Not English	0.45	0.47	0.45	0.49	0.05 (0.22)	0.08 (0.10)
Learning Disability	0.32	0.07	0.24	0.09	-0.67 (0.00)	-0.40 (0.00)
Homeless	0.06	0.02	0.06	0.04	-0.23 (0.00)	-0.13 (0.03)

Student N	973	1,669	659	936
School N	23	23	23	23

Note. Column 1 presents means for students assigned to TSI in schools that adopted Edmentum products during the 2024–25 school year. Column 2 presents means for students not assigned to TSI in the same Edmentum schools. Columns 3 and 4 restrict columns 1 and 2 to students whose diagnostic scores fall within the covariate-adjusted MSE-optimal bandwidth of the Spring 2024–25 level RD (65.80 scale-score points in reading, 72.40 in mathematics) around the grade-specific eligibility cutoff. Columns 5 and 6 present standardized mean differences — between the full assigned and non-assigned groups (5) and between the assigned and non-assigned groups within the bandwidth (6) — with p-values in parentheses. Standardized differences are calculated as the difference in group means divided by the pooled standard deviation of the full sample, so the within-bandwidth differences are on the same scale as the full-sample differences.

Table 3

The Level RD at the TSI Eligibility Cutoff and Its Exact Decomposition into Baseline and Growth Discontinuities (95% CI in Brackets)

	Reading N=912	Mathematics N=1,358
Level RD, Spring 2024-25 (research question 1)	0.306 [0.174 - 0.815]	-0.166 [-0.494 - 0.117]
Baseline discontinuity, Spring 2023-24	0.196 [-0.149 - 0.629]	-0.160 [-0.583 - -0.055]
Growth discontinuity (difference-in-discontinuities, research question 2)	0.110 [-0.107 - 0.616]	-0.006 [-0.139 - 0.399]

All estimates are standardized to the Spring 2023-24 control mean and standard deviation within the covariate-adjusted MSE-optimal bandwidth and are computed on the balanced sample of students observed in both Spring 2023-24 and Spring 2024-25 (N shown in the column headers). All rows are covariate-adjusted local-linear (triangular kernel) sharp RD estimates at the fixed bandwidth with robust bias-corrected 95% CIs. Because the RD estimator is linear in the outcome at a fixed bandwidth, row 1 equals row 2 plus row 3 exactly: the endline level discontinuity decomposes into the gap that predates the intervention (row 2) and the discontinuity in Spring-to-Spring achievement growth (row 3, the difference-in-discontinuities, which identifies the ITT effect of assignment at the cutoff under the time-invariance assumption tested in the stacked-RD and functional-form stability checks).

Table 4

The Bridge: The Bandwidth-Restricted Two-Period IPW DiD and Its Exact RD Twin (95% CI in Brackets)

	Reading	Mathematics
	N=912	N=1,358
Two-period IPW DiD (cluster-robust CI)	0.179 [0.009 - 0.349]	-0.120 [-0.268 - 0.028]
The same estimate as an RD (robust bias-corrected CI)	0.179 [-0.193 - 0.288]	-0.120 [-0.307 - 0.097]

Both rows report the same weighted least squares coefficient, computed on the balanced near-cutoff sample of students observed in both Spring 2023-24 and Spring 2024-25 (N shown in the column headers) and standardized to the Spring 2023-24 control mean and standard deviation within the bandwidth. Row 1 is the two-period IPW change-score DiD (student fixed effects) with its native cluster-robust CI. Row 2 re-computes the identical point estimate as a sharp RD on the change score with a local-constant fit, a uniform kernel, IPW unit weights, and the fixed bandwidth, reported with rdrobust's robust bias-corrected CI, which targets the limit at the cutoff, prices the local approximation error near the boundary, and is centered on the bias-corrected rather than the conventional estimate. The two rows agree to machine

precision by construction, so any disagreement in statistical significance between them isolates the different inferential standards of the two designs rather than any difference in specification.

Table 5

ITT Effect of TSI Assignment for the Full Assigned Population: Two-Period IPW DiD (95% CI in Brackets)

	Reading	Mathematics
	N=2,289	N=2,211
Two-period IPW DiD, all assigned students (research question 3)	0.251	-0.120
	[0.135 - 0.367]	[-0.234 - -0.007]

The two-period IPW change-score DiD (Spring 2023-24 to Spring 2024-25, student fixed effects, ATT inverse probability weights from the difference-score propensity model, cluster-robust 95% CIs) estimated on the full within-school analytic sample: TSI-assigned students and all non-assigned students in Edmentum schools, restricted to the matched sample and common support and balanced on the two comparison periods (N shown in the column headers). Outcomes are standardized to the Spring 2023-24 control mean and standard deviation of this sample. The estimator is identical to the bandwidth-restricted bridge specification; the only change is the population, so the difference between this estimate and the bridge estimate reflects the population over which the effect of assignment is averaged rather than any change in design.

Table 6

ITT Effects of TSI Assignment on Student Achievement: Six-Period Event Study, TSI Assigned vs. All Non-Assigned

	(1)	(2)
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	Reading	Mathematics
Fall 2324	-0.0453 (0.0519)	0.0742 (0.0417)
Winter 2324	0.0241 (0.0424)	0.0522 (0.0390)
Spring 2324	0 (.)	0 (.)
Fall 2425	0.0433 (0.0510)	0.0439 (0.0498)
Winter 2425	0.156*** (0.0441)	-0.0303 (0.0425)
Spring 2425	0.234*** (0.0471)	-0.120** (0.0455)
N	2584	2567
Pooled Estimate	0.195	-0.0752
Pooled Estimate SE	0.0422	0.0390
Pooled Estimate P-value	0.00000404	0.0537
Pre Trends Joint F	1.006	1.074
Pre Trends Joint F df	3	3
Pre Trends Joint F pval	0.389	0.359

Standard errors in parentheses * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. The dependent variable is the Renaissance Star assessment score, standardized using the mean and standard deviation of the control group in the reference period (Spring 2023–24). Coefficients are therefore interpreted as standard deviation units relative to the control group baseline. Columns present event study estimates for students assigned to TSI versus all non-assigned students

in the same Edmentum-adopting schools, for the reading and mathematics analytic samples respectively. The treatment periods are Winter 2024–25 and Spring 2024–25. Spring 2023–24 serves as the omitted reference category; the remaining testing periods are used to assess pre-trends. Event study coefficients are estimated using an inverse probability–weighted two-way fixed effects model with student and testing period fixed effects, with cluster-robust standard errors computed at the student level. Pooled estimates are reported below the event study coefficients and represent the simple average of the two post-treatment coefficients (Winter 2024–25 and Spring 2024–25), with standard errors calculated using the delta method. A joint F-test of pre-trends across the Fall 2023–24, Winter 2023–24, and Fall 2024–25 periods is presented below the pooled estimates. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 7

Four Estimates of One Effect: From the Level Discontinuity at the Cutoff to the Full-Population Event Study (95% CI in Brackets)

	Reading	Mathematics
Level RD at the cutoff, Spring 2024-25 (RQ1)	0.306 [0.174 - 0.815]	-0.166 [-0.494 - 0.117]
Baseline discontinuity, Spring 2023-24	0.196 [-0.149 - 0.629]	-0.160 [-0.583 - -0.055]
Growth discontinuity at the cutoff (RQ2)	0.110 [-0.107 - 0.616]	-0.006 [-0.139 - 0.399]
Bridge: two-period IPW DiD within the bandwidth (cluster-robust CI)	0.179 [0.009 - 0.349]	-0.120 [-0.268 - 0.028]
Bridge as RD (robust bias-corrected CI)	0.179 [-0.193 - 0.288]	-0.120 [-0.307 - 0.097]

Two-period IPW DiD, all assigned students (RQ3)	0.251	-0.120
	[0.135 - 0.367]	[-0.234 - -0.007]
Six-period event study, Spring 2024-25 (RQ4)	0.234	-0.120
	[0.142 - 0.327]	[-0.209 - -0.031]
Six-period event study, pooled post-intervention (RQ4)	0.195	-0.075
	[0.112 - 0.278]	[-0.152 - 0.001]

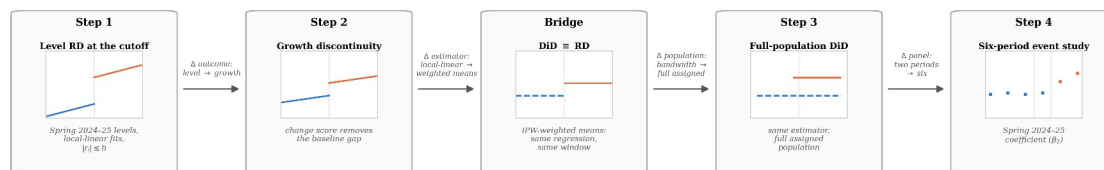
Each step changes exactly one element relative to the step before it. Rows 1-3 are covariate-adjusted local-linear (triangular kernel) sharp RD estimates at the fixed covariate-adjusted MSE-optimal bandwidth with robust bias-corrected CIs, computed on the balanced near-cutoff sample (N = 912 reading, 1,358 mathematics) and standardized within the bandwidth; row 1 equals row 2 plus row 3 exactly. Rows 4 and 5 are the same weighted least squares coefficient -- the bandwidth-restricted two-period IPW DiD, algebraically identical to a sharp RD on the change score with a local-constant fit, uniform kernel, and IPW unit weights -- reported under cluster-robust and robust bias-corrected inference respectively. Row 6 holds the estimator fixed and extends the population to the full within-school analytic sample (N = 2,289 reading, 2,211 mathematics; standardized to that sample's Spring 2023-24 control moments), so the movement from row 4 to row 6 reflects the population over which the effect is averaged, not the design. Rows 7 and 8 extend the panel to all six testing periods (N = 2,584 reading, 2,567 mathematics, unbalanced): the Spring 2024-25 coefficient and the pooled post-intervention average from the six-period IPW event study, with cluster-robust CIs. Rows 1-5 and rows 6-8 are standardized to different control-group moments (within-bandwidth vs. full-sample), so small level differences across those blocks partly reflect scale.

Figures

Figure 1.

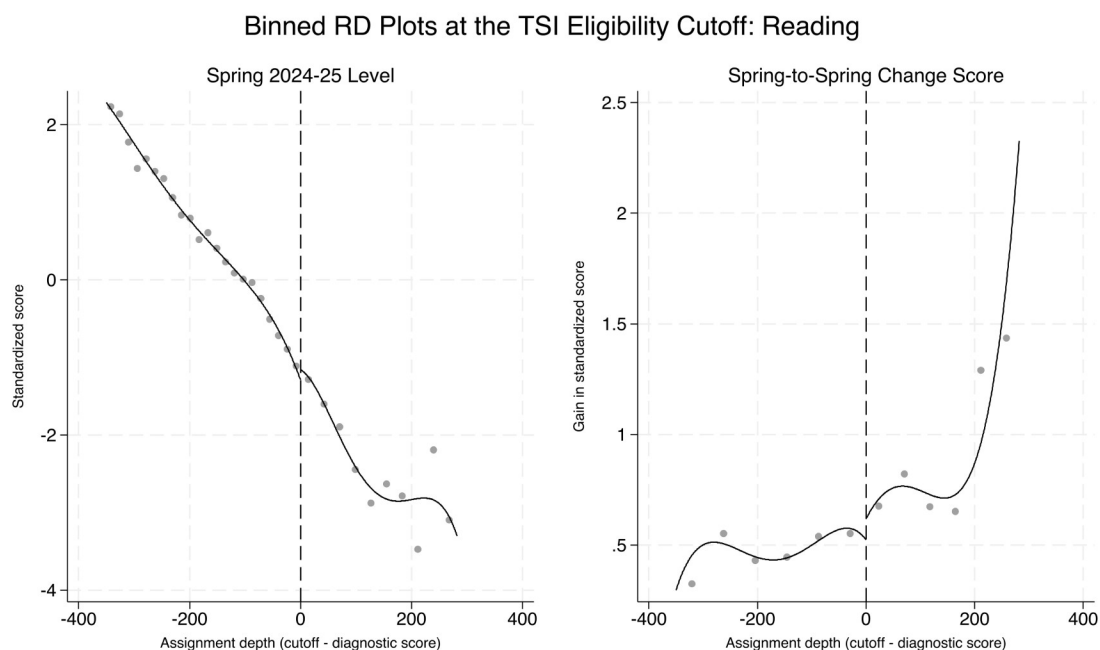
Four Estimates of One Effect: From the Cutoff to the Full Assigned Population

Each step changes exactly one element of the design.



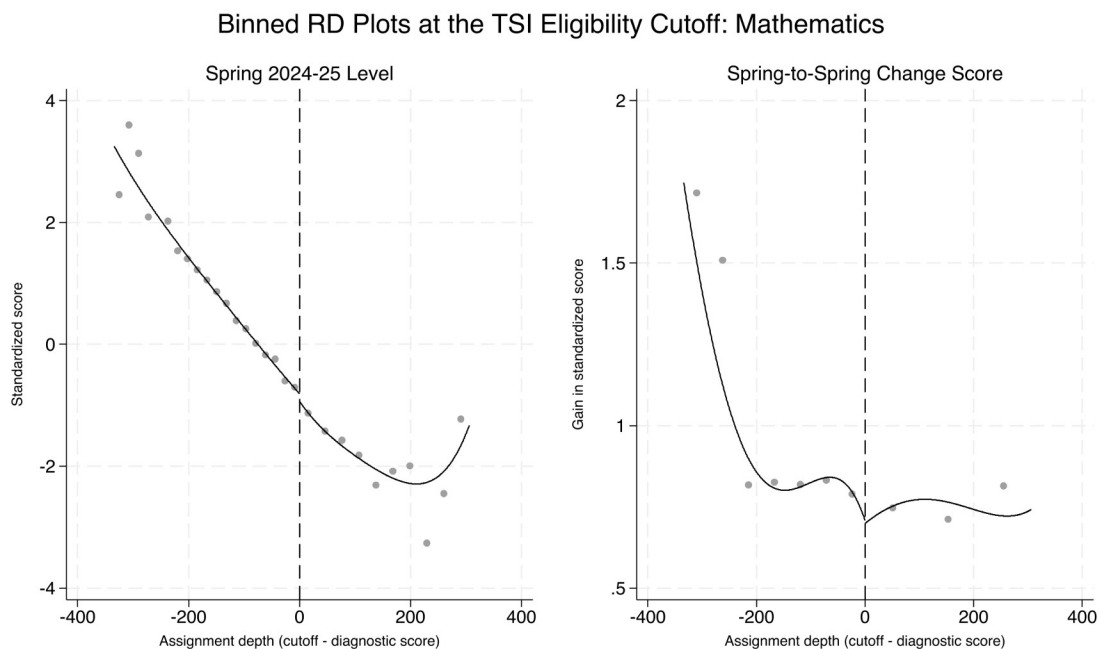
Note. The figure presents the four steps as five panels in one row, one per estimate plus the bridge. Each panel shows a stylized rendering of its design: the sharp RD in Spring 2024–25 levels; the growth discontinuity on the change score; the bridge, in which IPW-weighted means within the same bandwidth reproduce the DiD estimator; the full-population DiD over the full assigned range; and the six-period event study. The italic labels on the arrows between panels name the single element of the design that changes at each step: the outcome (levels to growth), the estimator (local-linear fits to weighted means), the population (bandwidth to full assigned), and the panel (two periods to six).

Figure 2.



Note. The left panel plots binned means of the Spring 2024–25 standardized score against assignment depth (the grade-specific eligibility cutoff minus the student's Exact Path Insights diagnostic score, so students assigned to TSI --- those scoring at or below the cutoff --- have non-negative values); the right panel repeats the construction for the Spring-to-Spring change score on the balanced sample of students observed in both Spring 2023–24 and Spring 2024–25. Bins are evenly spaced with the IMSE-optimal number of bins selected by `rdplot`, and solid lines are fourth-order global polynomial fits estimated separately on each side of the cutoff (Calonico, Cattaneo, & Titiunik, 2014). The sample is students in TSI-offering Edmentum schools with a diagnostic score. Level scores are standardized to the Spring 2024–25 non-assigned mean and standard deviation; change scores are differences of scores standardized to the Spring 2023–24 non-assigned mean and standard deviation. A visible jump at zero in the left panel corresponds to the level discontinuity reported in [Table 3](#); a jump in the right panel corresponds to the growth discontinuity (difference-in-discontinuities) reported in the same table.

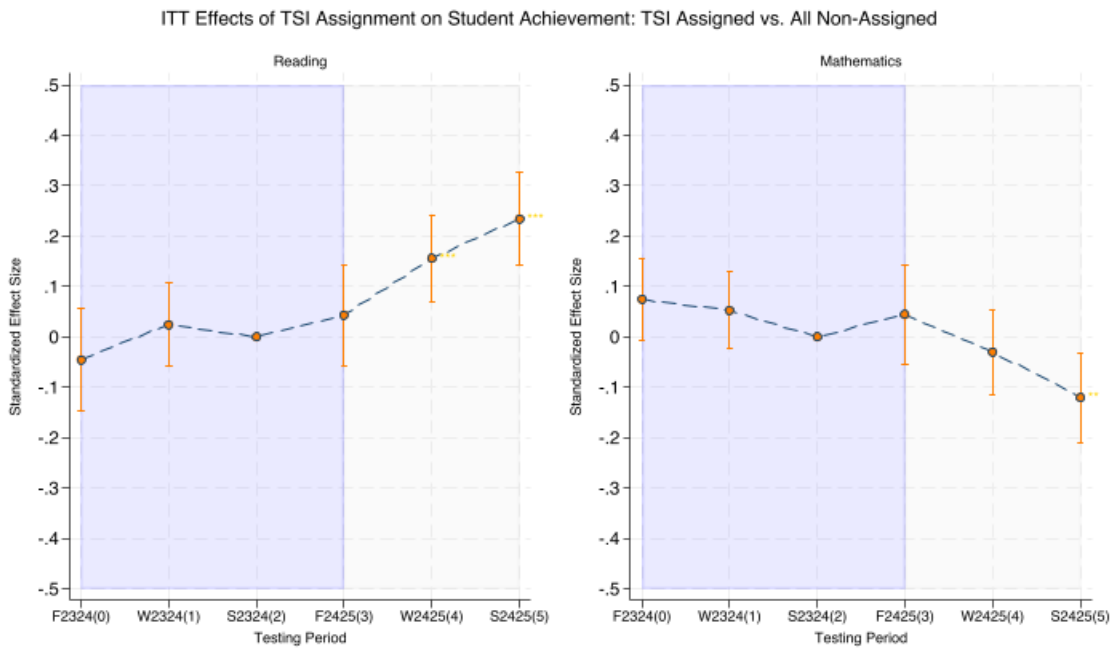
Figure 3.



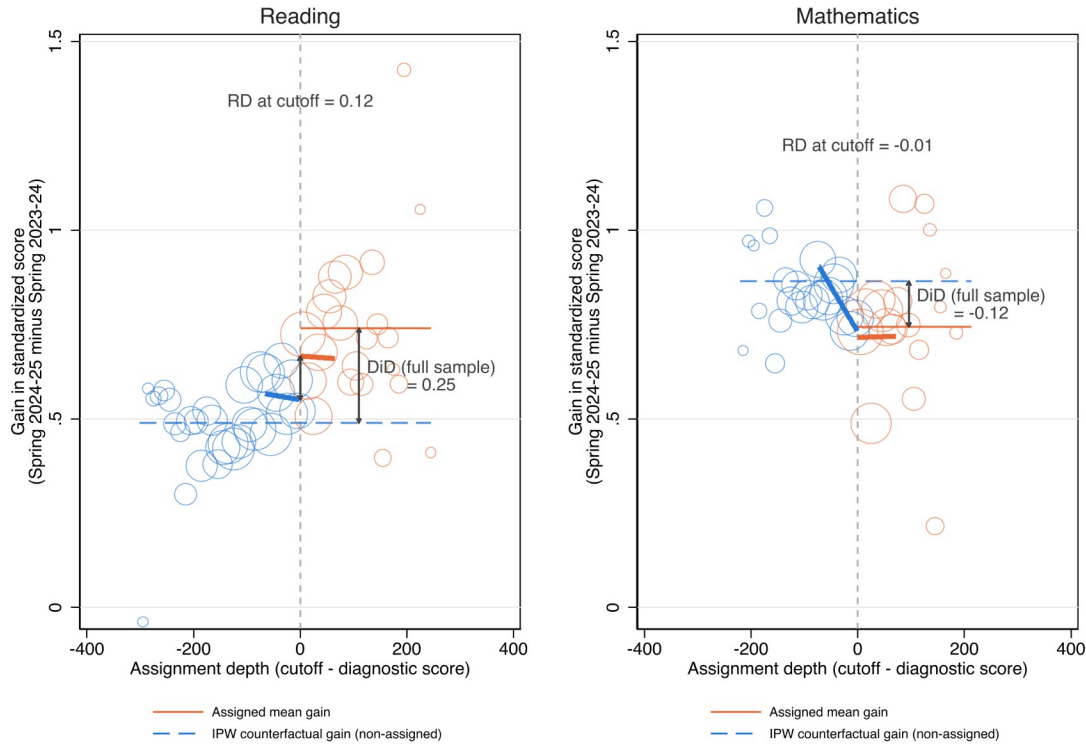
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Spring-to-Spring change score on the balanced sample of students observed in both Spring 2023–24 and Spring 2024–25. Bins are evenly spaced with the IMSE-optimal number of bins selected by `rdplot`, and solid lines are fourth-order global polynomial fits estimated separately on each side of the cutoff (Calonico, Cattaneo, & Titiunik, 2014). The sample is students in TSI-offering Edmentum schools with a diagnostic score. Level scores are standardized to the Spring 2024–25 non-assigned mean and standard deviation; change scores are differences of scores standardized to the Spring 2023–24 non-assigned mean and standard deviation. A visible jump at zero in the left panel corresponds to the level discontinuity reported in Table 3; a jump in the right panel corresponds to the growth discontinuity (difference-in-discontinuities) reported in the same table.

Figure 4.



Note. The figure plots event-study coefficients for the effect of TSI assignment on standardized Renaissance Star scores in the generalization sample, for reading (left) and mathematics (right). Coefficients are from an inverse probability-weighted two-way fixed effects model with student and testing-period fixed effects, normalized to Spring 2023–24, with 95% cluster-robust confidence intervals. The shaded region marks the post-intervention periods (Winter and Spring 2024–25). Stars denote significance: * $p < .05$, ** $p < .01$, *** $p < .001$.

Figure 5.

Note. The figure plots binned mean gains in the standardized score between Spring 2023–24 and Spring 2024–25 against the running variable over its full range, for the generalization sample restricted to the matched sample and common support and balanced on the two comparison springs — the sample of the full-population two-period estimate (Table 5). Marker area is proportional to the number of students in each width-10 bin; bins with fewer than five students are omitted. Thick line segments are local-linear (triangular kernel) fits estimated separately on each side of the cutoff within the fixed covariate-adjusted MSE-optimal bandwidth; their vertical gap at zero is the change-score RD at the cutoff. Horizontal reference lines mark the mean gain of assigned students (solid) and the IPW-reweighted mean gain of non-assigned students (dashed); their vertical gap equals the two-period IPW DiD for the full assigned population (Table 5) to machine precision. All quantities are standardized to the sample's Spring 2023–24 control-group mean and standard deviation, the scale of Tables 5 and 6. The contrast between the gap at the cutoff and the gap between the horizontal lines is the generalization result: where they differ, the effect for the full assigned population differs from the effect at the eligibility margin.